In Silico Closed-Loop Metabolic Gating of Glycolytic-to-Oxidative Transition in *Ex Vivo* CAR-T Perfusion Bioreactors Guided by Structural Bioinformatics and Optical Redox Observability

O. Satz[1,*]

[1]Department of Bioengineering and Embedded Control Systems, MitoRestorer Research Group

[*]Corresponding author: `osatz@mitorestorer.org` | Repository: https://github.com/deesatzed/mitorestorer

November 2026 (MitoRestorer Series, Paper II; Computational Study — MIASE Seed: `20261111`)

Abstract

Durable clinical responses in adoptive chimeric antigen receptor (CAR) T-cell therapy correlate with infusing stem-like memory T cells (T_{SCM}, $CD45RA^+CD62L^+CCR7^+TCF7^+$) possessing high mitochondrial spare respiratory capacity (SRC), yet 168-hour *ex vivo* perfusion expansion under continuous high glucose and interleukin-2 (IL-2) enforces a Warburg glycolytic program that accumulates depolarized mitochondria and TOX^+ epigenetic exhaustion. Here we present a purely *in silico* systems-and-control investigation—combining structural bioinformatics, eight-state nonlinear ordinary differential equation (ODE) modeling, discrete-adjoint optimal control, nonlinear observability analysis, and host-emulated fixed-point firmware verification—to design falsifiable closed-loop metabolic gating protocols prior to wet-lab execution. Heavy-atom analysis of experimental crystal structures confirms druggable allosteric pockets on pyruvate dehydrogenase kinase 1 (`PDK1`, `PDB: 2Q8G`, 1.90 Å; lipoyl-pocket `Ser75` hydrogen bond at 2.50 Å to `AZD7545`) and sirtuin 1 (`SIRT1`, `PDB: 4ZZJ`, 2.74 Å; `Glu230` allosteric hinge at 3.47 Å), whereas full-length AlphaFold v6 models of downstream effectors `FOXO1` (655 AA, mean pLDDT = 50.96) and `PGC-1`α (798 AA, mean pLDDT = 52.74) exhibit single structured nucleic-acid-binding modules flanked by intrinsically disordered post-translational modification regions (9/9 `Akt`/`SIRT1` sites on `FOXO1` and 39/47 lysines on `PGC-1`α have pLDDT < 70). Building on the nondimensional selection framework of Paper I, our simulated two-stage **Bioreactor-BCR** controller separates Stage-1 mitophagy-assisted glycolytic blastogenesis (u_1) from Stage-2 pyruvate dehydrogenase reactivation and NAD^+/`SIRT1`-coupled oxidative consolidation (u_2), yielding 62.80× simulated fold expansion, 73.20% T_{SCM}, SRC = 4.87×, and $h(168\text{ h}) = 6.73\%$ on the nominal model, closely tracking the 200-variable discrete-adjoint Pontryagin Pareto frontier (gradient error 9.93×10^{-11}, 5-start cost spread 1.12%). Across $N = 200$ simulated heterogeneous oncology apheresis donors with ±16%–36% unmodeled kinetic variation, uncalibrated four-channel supernatant sensing (`Lac:Pyr`, specific oxygen uptake, optical density, and pH) fails under Warburg-yield confounding (37.5% 5-criteria pass rate) and suffers a 342× Observability Gramian conditioning collapse ($\kappa = 5.52 \times 10^3 \rightarrow 1.89 \times 10^6$) upon enzymatic electrode biofouling. Augmenting the cassette observer with a fouling-immune, label-free autofluorescence optical redox ratio (`ORR`) channel restores well-conditioned state estimation ($\hat{h}$ RMSE 1.83%), raising the simulated cohort pass rate from 48.5% (best fixed 68 h timer) to 55.0% (67.5% full-information donor oracle ceiling) and retaining 54.0% under 100% amperometric `Lac:Pyr` biofouling via normalized innovation squared (NIS > 13.5) optical handover. Finally, bare-metal C99 firmware with `Q16.16` states, `Q8.24` slopes, and 64-bit $\Sigma\Delta$ remainder tracking fits in 3,920 B (47.85%) of single-bank L1 DTCM on a host-emulated `RV32IMF_Zve32x` neural processing unit (0.0078%–0.0114% `RK4` error across 500–8,000 steps; 3.108 mW active burst power) and rescues simulated perfusion-line acidosis (min pH 6.14 → 6.91) without discontinuous plant-state resets.

1 Introduction

1.1 Bioenergetic Determinants of CAR-T Persistence

In autologous CD19- and BCMA-directed chimeric antigen receptor (CAR) T-cell manufacturing, clinical durability in chronic lymphocytic leukemia, B-cell acute lymphoblastic leukemia, and non-Hodgkin lymphoma is governed less by total infused cell count than by the proportion of early stem-cell memory T lymphocytes (T_{SCM}, $CD8^+CD45RA^+CD62L^+CCR7^+TCF7^+$) and their mitochondrial spare respiratory capacity (SRC) [1, 2]. Transcriptomic and epigenetic studies have established two

converging mechanistic axes underlying this phenotype: first, nuclear retention of Forkhead box protein O1 (FOXO1) acts as an essential master regulator of CAR-T memory programming and metabolic fitness while opposing TOX^+ terminal exhaustion [3, 4, 5]; second, selective clearance of depolarized, fragmented mitochondria via PINK1/Parkin-dependent mitophagy (stimulated *ex vivo* or *in vivo* by urolithin A) preserves fused cristae networks and expands T_{SCM} subsets [6, 7, 8].

1.2 The Ex Vivo Proliferation–Exhaustion Conflict

Automated closed-cassette perfusion bioreactors (such as hollow-fiber or gas-permeable membrane cassettes) activate patient apheresis T cells via anti-CD3/CD28 stimulation over a 7-day (168 h) expansion cycle. This manufacturing window imposes a well-known dynamical tradeoff between clonal blastogenesis and bioenergetic stemness [9, 2]:

1. **Unphased Aerobic Glycolysis (Warburg Lock):** Continuous exposure to high glucose (25 mM) and high-dose IL-2 (300 IU/mL) drives mTORC1, Akt, and HIF-1α signaling. HIF-1α upregulates pyruvate dehydrogenase kinase 1 (PDK1) [10], which phosphorylates the pyruvate dehydrogenase (PDH) E1α subunit, shunting pyruvate to lactate ($\mathrm{Lac:Pyr} > 25$) and acidifying the perfusate. Concurrently, Akt phosphorylates FOXO1 at Thr24, Ser256, and Ser319, triggering 14-3-3 chaperone binding and nuclear export [3, 4], while mTORC1 suppresses ULK1-initiated mitophagy, allowing depolarized organelles (D) and TOX^+ chromatin remodeling to accumulate [9, 5].

2. **Premature Oxidative Forced Switching (Dose Failure):** Conversely, forcing oxidative phosphorylation or restricting glycolytic flux immediately upon CD3/CD28 activation ($t = 0$ h) deprives dividing lymphoblasts of glycolytic ribonucleotide, serine, and lipid precursors required during early log-phase expansion, risking failure to reach minimum clinical release doses ($\geq 50\times$ viable fold expansion).

3. **Oncology Apheresis Heterogeneity and Sensor Drift:** Unlike healthy-donor research runs, heavily pretreated oncology patients enter apheresis with wide baseline dispersion in T-cell mitochondrial depolarization ($h_0 \in [0.20, 0.62]$) and division kinetics [1]. Consequently, a fixed open-loop media-switch timer tuned to a nominal donor risks switching slow-clearing donors prematurely or holding rapid-proliferating donors in prolonged glycolysis. Yet closed-loop gating using inline enzymatic supernatant sensors faces its own bioprocess hazard: amperometric lactate/pyruvate electrodes undergo progressive protein biofouling over multi-day perfusion.

1.3 Epistemic Scope and Computational Contributions

Disclosure of Computational Scope: This study is strictly a theoretical, structural-bioinformatic, and *in silico* control-engineering investigation. *No wet-lab, in vitro, ex vivo, or in vivo biological experiments were performed.* All cell-expansion trajectories, heterogeneous patient-cohort statistics, biofouling fault responses, and embedded processor benchmarks are generated by deterministic mathematical models and host-emulated firmware designed to formulate quantitative, falsifiable predictions for subsequent experimental testing (§7).

Extending the nondimensional staged heteroplasmy control theory of Paper I [11] to proliferating cell populations coupled to extracellular bioreactor mass transfer, this paper makes four specific contributions:

1. **Unified 8-State Perfusion Cassette ODE with 1-to-1 Code Parity (§2):** We formulate an eight-state bioreactor model coupling intracellular mitochondrial subpopulations (W, D), clonal cell fold expansion (N_{cells}), supernatant lactate-to-pyruvate ratio (R_{LP}), nuclear FOXO1 fraction (F_{nuc}), cristae-linked SRC (S_{SRC}), epigenetic exhaustion (E_{TOX}), and perfusate pH, with every closed-form auxiliary equation verified identically across FP64 Python, smooth discrete-adjoint optimization, and bare-metal C99 fixed-point code.

2. **Atomic Contact & AlphaFold v6 Disorder Audit Guiding Actuator Selection (§3):** By auditing heavy-atom ligand contacts in PDB: 2Q8G (PDK1–AZD7545) and PDB: 4ZZJ (SIRT1–STAC–carba-NAD^+) alongside AlphaFold Database v6 pLDDT and Predicted Aligned Error (PAE) matrices for full-length FOXO1 and PGC-1α, we quantify why disordered regulatory PTM tails lack preformed small-molecule pockets and why upstream enzymatic actuation—paired with explicit controls for known STAC chemical-probe artifacts [12, 13]—provides a structurally grounded intervention strategy.

3. **Discrete-Adjoint Pareto Frontier & Sensor Observability Under Donor Heterogeneity (§4–§5):** Solving a 200-variable discrete-adjoint optimal control problem (L-BFGS-B, 5-start spread 1.12%) maps the simulated proliferation–stemness Pareto frontier. Across $N = 200$ heterogeneous oncology donors, we prove via empirical Observability Gramian analysis why supernatant Lac:Pyr + respirometry alone fails under Warburg-rate heterogeneity and electrode biofouling ($\kappa = 1.89 \times 10^6$), and show that adding a fouling-immune inline autofluorescence optical redox ratio (ORR) channel [14] with NIS fault handover raises the 5-criteria cohort pass rate from 48.5% (best fixed 68 h timer) to 55.0% (54.0% under 100% Lac:Pyr fouling).

4. **Plant-Preserving Acidosis Flush & Host-Emulated `RV32IMF_Zve32x` Firmware (§6):** Eliminating unphysical state-reset assumptions, we verify a continuous-plant neutral-buffer bypass flush (u_{fl}) during simulated perfusion-line occlusion and demonstrate exact C99 `Q16.16/Q8.24` ΣΔ `RK4` execution within 3,920 B of L1 DTCM Bank 0.

2 Mathematical Model of the 168-Hour Perfusion Cassette

2.1 Eight-State Cassette Vector and Bounded Controls

We model an automated closed-cassette CAR-T perfusion culture over $t \in [0, T_{\max}]$ ($T_{\max} = 168.0$ h) via the state vector

$$\mathbf{x}(t) = \big[W,\, D,\, N_{\mathrm{cells}},\, R_{\mathrm{LP}},\, F_{\mathrm{nuc}},\, S_{\mathrm{SRC}},\, E_{\mathrm{TOX}},\, \mathrm{pH}\big]^{\top}, \tag{1}$$

where $W(t) \geq 0$ is normalized intact, fused-cristae wild-type mitochondrial mass per cell, $D(t) \geq 0$ is normalized depolarized/fragmented mitochondrial mass per cell, $N_{\mathrm{cells}}(t) \geq 1$ is viable clonal fold expansion ($N_{\mathrm{cells}}(0) = 1.0$), $R_{\mathrm{LP}}(t) > 0$ is the extracellular supernatant molar [Lactate]/[Pyruvate] ratio, $F_{\mathrm{nuc}}(t) \in [0, 1]$ is the nuclear-retained, transcriptionally active fraction of `FOXO1`, $S_{\mathrm{SRC}}(t) \geq 0$ is mitochondrial spare respiratory capacity (dimensionless multiple of basal oxygen consumption), $E_{\mathrm{TOX}}(t) \in [0, 1]$ is the fraction of cells committed to TOX^{+}PD-1^{+}TIM-3^{+} epigenetic exhaustion, and pH(t) is the cassette perfusate pH. Total normalized organelle mass is $M(t) = W(t) + D(t)$, with depolarized fraction (functional heteroplasmy) $h(t) = D(t)/M(t)$ and intact fraction $w_{\mathrm{frac}}(t) = 1 - h(t) = W(t)/M(t)$.

The cassette is actuated by three bounded inputs $\mathbf{u}(t) = [u_1(t), u_2(t), u_{\mathrm{fl}}(t)]^{\top} \in [0, 1]^3$:

- $u_1(t) \in [0, 1]$: **Stage-1 Mitophagy & Mild Glycolytic Restraint Perfusate (Media-A)**, modeling urolithin A (25 μM) mitophagy induction paired with moderate `PI3Kδ/Akt` restraint during high-glucose blastogenesis [7, 4];
- $u_2(t) \in [0, 1]$: **Stage-2 Oxidative Stemness Perfusate (Media-B)**, modeling iso-osmotic partial galactose substitution ($\alpha_{\mathrm{gal}} = 0.74$), allosteric `PDK1` lipoyl-pocket inhibition (`AZD7545`), NAD^{+} repletion / `SIRT1` activation, and homeostatic IL-7/IL-15 cytokine support [10, 15];
- $u_{\mathrm{fl}}(t) \in [0, 1]$: **Emergency Neutral-Buffer Bypass Flush Valve**, routing fresh pH$_{\mathrm{media}} = 7.36$ buffer across the hollow-fiber cartridge if main perfusion-line occlusion triggers acute lactic acidosis.

2.2 Closed-Form Auxiliary Bioenergetic Equations

To guarantee zero discrepancy between analytical expressions, `eval_cart8_rhs_vec` in Python, and `cart8_eval_rhs_q16` in C99, every intermediate rate is specified explicitly:

$$g_{\mathrm{glc}}(u_2) = 1 - \alpha_{\mathrm{gal}}\, u_2, \quad g_{\mathrm{pdh}}(u_2) = 0.15 + 0.85\, u_2, \tag{2}$$

$$v_{\mathrm{unmask}}(R_{\mathrm{LP}}) = \frac{K_{\mathrm{LP}}}{K_{\mathrm{LP}} + (R_{\mathrm{LP}} - R_{\mathrm{basal}})_{+}}, \tag{3}$$

$$\begin{aligned} k_{\mathrm{mito},D}(u_1, R_{\mathrm{LP}}) &= k_{\mathrm{mito},W} \\ &\quad + (k_{\mathrm{mito},D,\max} - k_{\mathrm{mito},W})\, u_1\, v_{\mathrm{unmask}}(R_{\mathrm{LP}}), \end{aligned} \tag{4}$$

$$\begin{aligned} \mathrm{ATP}_{\mathrm{ox}}(\mathbf{x}, u_2) &= (W + \eta_D D)(1 + \gamma_{\mathrm{pdh}} u_2) \\ &\quad \times \big(1 + \gamma_{\mathrm{src}}(S_{\mathrm{SRC}} - 1)_{+}\big), \end{aligned} \tag{5}$$

$$\begin{aligned} \mathrm{ATP}_{\mathrm{glyc}}(\mathbf{x}, \mathbf{u}) &= \kappa_{\mathrm{glyc}}\, g_{\mathrm{glc}}(u_2)\big(1 - c_{\mathrm{akt}} u_1 (1 - u_2)\big) \\ &\quad \times \frac{R_{\mathrm{LP}}}{R_{\mathrm{LP}} + K_{R,\mathrm{ATP}}}, \end{aligned} \tag{6}$$

$$\phi_{\mathrm{pH}}(\mathrm{pH}) = \frac{1}{1 + \big((7.02 - \mathrm{pH})_{+}/0.16\big)^2}, \tag{7}$$

$$\begin{aligned} \dot{e}_{\mathrm{load}}(\mathbf{x}) &= \big[1 + \beta_{\mathrm{LP}}(R_{\mathrm{LP}} - R_{\mathrm{basal}})_{+}/K_{\mathrm{LP}} + \beta_h h \\ &\quad + \beta_{\mathrm{pH}}(7.02 - \mathrm{pH})_{+}\big] / \max(0.26,\, W + \eta_D D), \end{aligned} \tag{8}$$

$$\mathrm{EP}(\mathbf{x}, \mathbf{u}) = \dot{e}_{\mathrm{load}}(\mathbf{x})\,(1 - c_1 u_1 + c_2 u_2)^2, \tag{9}$$

$$w_{\mathrm{func}}(\mathbf{x}) = W(1 - h),$$

$$\Phi_{\mathrm{ox}}(\mathbf{x}) = \frac{w_{\mathrm{func}}^2}{w_{\mathrm{func}}^2 + K_{W,\mathrm{div}}^2}. \tag{10}$$

Here $v_{\mathrm{unmask}}(R_{\mathrm{LP}})$ captures Warburg/`mTORC1` suppression of `ULK1/PINK1` mitophagic flux under excessive lactate:pyruvate accumulation, ϕ_{pH} captures severe proton-mediated inhibition of blastogenesis and nuclear import below pH 7.02, and Φ_{ox} requires intact cristae mass w_{func} to sustain rapid oxidative division in low-glucose Stage-2 media.

2.3 Coupled Governing ODEs and Readout Map

With total cellular ATP availability $\mathrm{ATP}_{\mathrm{tot}} = \mathrm{ATP}_{\mathrm{ox}} + \mathrm{ATP}_{\mathrm{glyc}}$ and damage conversion rate $d_{\mathrm{ros}}(\mathbf{x}, \mathbf{u}) =$

$k_{\mathrm{dmg}}\max(0,\mathrm{EP}-\theta_{\mathrm{GSH}})$, the eight-state ODE system is:

$$\dot{W} = \big(k_{\mathrm{bio},0} + (k_{\mathrm{bio,max}} - k_{\mathrm{bio},0})\,u_2 F_{\mathrm{nuc}}\big) W\Big(1-\frac{M}{K}\Big)_+ - \big(k_{\mathrm{mito},W} + d_{\mathrm{ros}}(\mathbf{x},\mathbf{u})\big)W, \tag{11}$$

$$\dot{D} = k_{\mathrm{rep,adv}}\big(k_{\mathrm{bio},0} + \xi_D(k_{\mathrm{bio,max}} - k_{\mathrm{bio},0})\,u_2\big) \times D\Big(1-\frac{M}{K}\Big)_+ - k_{\mathrm{mito},D}(u_1, R_{\mathrm{LP}})\,D + d_{\mathrm{ros}}(\mathbf{x},\mathbf{u})\,W, \tag{12}$$

$$\dot{N}_{\mathrm{cells}} = \Big[\mu_{\mathrm{glyc}} g_{\mathrm{glc}}(u_2)\big(1-0.08u_1(1-u_2)\big) + \mu_{\mathrm{ox}} u_2 g_{\mathrm{pdh}}(u_2)\Phi_{\mathrm{ox}}\Big] \times \phi_{\mathrm{pH}}\,\frac{\mathrm{ATP}_{\mathrm{tot}}}{\mathrm{ATP}_{\mathrm{tot}}+K_{\mathrm{ATP}}}\,N_{\mathrm{cells}}\Big(1-\frac{N_{\mathrm{cells}}}{N_{\mathrm{cap}}}\Big)_+, \tag{13}$$

$$\dot{R}_{\mathrm{LP}} = k_{\mathrm{Warburg}} g_{\mathrm{glc}}(u_2)\big(1-0.35u_1(1-u_2)\big)(1+\alpha_h h) - k_{\mathrm{PDH}}(1+\beta_{\mathrm{azd}}u_2)w_{\mathrm{frac}}R_{\mathrm{LP}} - (k_{\mathrm{perf,LP}} + k_{\mathrm{fl,LP}}u_{\mathrm{fl}})(R_{\mathrm{LP}} - R_{\mathrm{media}}), \tag{14}$$

$$\dot{F}_{\mathrm{nuc}} = k_{\mathrm{imp}}\big(f_0 + \alpha_{\mathrm{akt}}u_1 + \phi_{\mathrm{SIRT1}}u_2 g_{\mathrm{pdh}}(u_2)\big) \times \phi_{\mathrm{pH}}\,(1-F_{\mathrm{nuc}}) - k_{\mathrm{exp}}(1-0.52u_1)\big(1-0.85u_2 g_{\mathrm{pdh}}(u_2)\big) \times \big(1+\gamma_{\mathrm{acid}}(6.90-\mathrm{pH})_+\big)F_{\mathrm{nuc}}, \tag{15}$$

$$\dot{S}_{\mathrm{SRC}} = s_{\mathrm{basal}}(W+\eta_D D) - k_{\mathrm{src,dec}}\,S_{\mathrm{SRC}} + k_{\mathrm{cristae}} w_{\mathrm{func}}\big(1+\beta_{\mathrm{src}}u_2 F_{\mathrm{nuc}} g_{\mathrm{pdh}}(u_2)\big), \tag{16}$$

$$\dot{E}_{\mathrm{TOX}} = \Big[k_{\mathrm{exh}}(\mathrm{EP}-\theta_{\mathrm{exh}})_+ + k_{\mathrm{acid,exh}}(6.86-\mathrm{pH})_+ + k_{\mathrm{warb,exh}} g_{\mathrm{glc}}(u_2)(1-0.22u_1)\Big(1+0.70\frac{N_{\mathrm{cells}}}{N_{\mathrm{ref}}}\Big)\Big] \times (1-E_{\mathrm{TOX}}) - \big(r_{\mathrm{epig},0} + k_{\mathrm{rev,epig}}u_2 F_{\mathrm{nuc}} g_{\mathrm{pdh}}(u_2)\big)E_{\mathrm{TOX}}, \tag{17}$$

$$\dot{\mathrm{pH}} = (k_{\mathrm{perf},0} + k_{\mathrm{perf,fl}}u_{\mathrm{fl}})(\mathrm{pH}_{\mathrm{media}} - \mathrm{pH}) - k_{\mathrm{acid}} g_{\mathrm{glc}}(u_2)(R_{\mathrm{LP}} - R_{\mathrm{buf}})_+ \frac{N_{\mathrm{cells}}}{N_{\mathrm{ref}}}, \tag{18}$$

where $(z)_+ \equiv \max(0,z)$. The terminal stem-like memory CAR-T fraction (%) is evaluated via the algebraic phenotypic readout

$$\mathrm{T}_{\mathrm{SCM}}(\mathbf{x}) = 100\,F_{\mathrm{nuc}}\,(1-E_{\mathrm{TOX}})\,\frac{S_{\mathrm{SRC}}}{S_{\mathrm{SRC}}+K_{\mathrm{SRC,mem}}}, \tag{19}$$

reflecting the joint requirement of nuclear FOXO1 retention (F_{nuc}), absence of TOX$^+$ epigenetic scarring ($1-E_{\mathrm{TOX}}$), and bioenergetic cristae reserve (S_{SRC}) [3, 4, 2].

Proposition 1 (Stage-1 Quotient Selection) *During Stage-1 mitophagy-primed expansion ($u_1 = 1, u_2 = 0$) whenever $\mathrm{EP}(\mathbf{x},\mathbf{u}) \le \theta_{\mathrm{GSH}}$ so $d_{\mathrm{ros}} = 0$, differentiating $h = D/(W+D)$ along (11)–(12) yields the exact Riccati quotient law of Paper I [11]:*

$$\dot{h}(t) = -\lambda_{\mathrm{CART}}(t)\,h(t)\big(1-h(t)\big), \tag{20}$$

with time-varying selective clearance rate

$$\lambda_{\mathrm{CART}}(t) = \big(k_{\mathrm{mito},D}(1, R_{\mathrm{LP}}(t)) - k_{\mathrm{mito},W}\big) - (k_{\mathrm{rep,adv}}-1)k_{\mathrm{bio},0}\Big(1-\frac{M(t)}{K}\Big)_+. \tag{21}$$

Consequently, in terms of the accumulated-selection coordinate $s(t) = \int_0^t \lambda_{\mathrm{CART}}(q)\,dq$, intracellular depolarization evolves exactly as $h(s) = [1 + \frac{1-h_0}{h_0}e^s]^{-1}$, reaching any target $h_{\mathrm{gate}} \in (0, h_0)$ at $s^* = \ln\big(\frac{h_0(1-h_{\mathrm{gate}})}{h_{\mathrm{gate}}(1-h_0)}\big)$.

Notice in (15)–(17) why *co-actuation* of PDK1 inhibition (g_{pdh}) and SIRT1/NAD$^+$ activation is required in Stage 2: without PDK1 inhibition ($u_{\mathrm{azd}} = 0 \implies g_{\mathrm{pdh}} = 0.15$), pyruvate remains blocked from the TCA cycle, collapsing both oxidative proliferation $\mu_{\mathrm{ox}} g_{\mathrm{pdh}} \Phi_{\mathrm{ox}}$ and NAD$^+$-coupled FOXO1 nuclear retention; conversely, without SIRT1 activation ($u_{\mathrm{sirt}} = 0$), wild-type biogenesis remains at basal $k_{\mathrm{bio},0}$ while depolarized organelles expand with replicative advantage $k_{\mathrm{rep,adv}} = 1.22$ via $\xi_D u_2$, causing $h(t)$ to rebound during Stage 2. All nominal parameters, initial conditions $\mathbf{x}(0) = [0.54, 0.42, 1.00, 22.0, 0.28, 1.30, 0.08, 7.32]^\top$ ($h_0 = 43.75\%$), and explicit provenance tags—[**M**] mechanistic/biophysical literature scale, [**F**] calibrated to published T-cell dynamic range, [**A**] operational cassette hardware parameter—are documented in Table 1.

3 Structural Bioinformatics Guiding Actuator Selection

A central design question for Stage-2 metabolic consolidation is why pharmacological actuation must target upstream metabolic kinases and deacetylases (PDK1 and SIRT1/NAD$^+$) rather than attempting direct small-molecule pocket activation of the master stemness transcription factors FOXO1 and PGC-1α. To ground our actuator parameterization in atomic coordinates, we audited experimental macromolecular crystallographic information files (.cif) from the Protein Data Bank alongside full-length AlphaFold Database v6 models and $N \times N$ Predicted Aligned Error (PAE) matrices [17, 18] (Figure 1, Table 2).

3.1 Druggable Allosteric Pockets in PDK1 (2Q8G) and SIRT1 (4ZZJ)

Coordinate inspection of human PDK1 co-crystallized with the trifluoropropanamide inhibitor AZD7545 (PDB: 2Q8G, Chain A, 368 resolved residues spanning 41–423 at 1.90 Å resolution [10]) confirms that AZD7545 (AZX, 31 heavy atoms) occupies the hydrophobic N-terminal lipoyl-binding (L2) pocket rather than the C-terminal ATP-binding cleft. Twelve residues form heavy-atom contacts $\le$ 4.0 Å to AZD7545 (Figure 1A): a short 2.50 Å hydrogen bond between Ser75 (Oγ) and the

Table 1: **Complete Parameterization and Provenance of the 8-State *In Silico* CAR-T Perfusion Bioreactor Model.** Every value matches `NOMINAL_PARAMS` in `simulator/generate_paper2_verified_benchmarks.py` and `build/coralnpu/cart_bioreactor_hil_testbench.c` 1-to-1. Provenance tags: **[M]** primary literature scale; **[F]** calibrated phenomenological coupling; **[A]** operational bioreactor/cassette specification.

Subsystem	Parameter(s)	Nominal Value(s)	Units	Tag	Biophysical Role & Literature Basis
Mitochondrial (W, D)	K, $k_{\mathrm{mito},W}$, $k_{\mathrm{bio},0}$	1.40, 0.008, 0.028	norm., h^{-1}, h^{-1}	[M/F]	Organelle cap & basal turnover [2, 11]
Mitophagy & Biogen.	$k_{\mathrm{mito},D,\max}$, $k_{\mathrm{bio},\max}$, $k_{\mathrm{rep,adv}}$, ξ_D	0.068, 0.086, 1.22, 1.00	h^{-1}, h^{-1}, –, –	[F]	Urolithin A clearance & replicative bias [7, 16]
Warburg Masking	R_{basal}, K_{LP}, η_D	8.0, 22.0, 0.22	molar ratio, –, –	[F]	`mTORC1/Lac:Pyr` masking of `ULK1`; D OXPHOS efficiency
Energetic Stress	γ_{pdh}, γ_{src}, α_{gal}, κ_{glyc}, c_{akt}, $K_{R,\mathrm{ATP}}$	0.72, 0.22, 0.74, 1.40, 0.08, 8.0	dimensionless	[F]	OXPHOS/glycolytic ATP & galactose shift [9]
ROS Damage Gate	β_{LP}, β_h, β_{pH}, c_1, c_2, θ_{GSH}, k_{dmg}	0.40, 0.88, 1.90, 0.24, 0.06, 1.38, 0.0012	–, h^{-1}	[F]	Threshold ROS conversion $W \to D$ above θ_{GSH} [11]
Clonal Division N	μ_{glyc}, μ_{ox}, K_{ATP}, $K_{W,\mathrm{div}}$, N_{cap}	0.062, 0.0185, 0.52, 0.56, 85.0	h^{-1}, –, fold	[M/A]	Lymphoblast doubling ($T_d \approx 11$–15 h) & cassette cap [1]
Supernatant R_{LP}	k_{Warburg}, α_h, k_{PDH}, β_{azd}, R_{media}	3.00, 0.46, 0.088, 2.80, 5.0	h^{-1}, –, ratio	[M/F]	Aerobic glycolysis vs. `AZD7545` `PDH` flux [10]
`FOXO1` Shuttle F_{nuc}	k_{imp}, f_0, α_{akt}, ϕ_{SIRT1}, k_{exp}, γ_{acid}	0.056, 0.30, 0.30, 1.18, 0.032, 7.5	h^{-1}, –	[M/F]	`Akt` 14-3-3 export vs. `SIRT1` retention [3, 4]
Cristae SRC	s_{basal}, k_{cristae}, β_{src}, $k_{\mathrm{src,dec}}$, $K_{\mathrm{SRC,mem}}$	0.044, 0.074, 1.32, 0.048, 0.68	h^{-1}, –	[M/F]	Cristae remodeling & $\mathrm{T_{SCM}}$ saturation [2]
Exhaustion E_{TOX}	k_{exh}, θ_{exh}, $k_{\mathrm{warb,exh}}$, $k_{\mathrm{acid,exh}}$, $r_{\mathrm{epig},0}$, $k_{\mathrm{rev,epig}}$	0.0038, 1.36, 0.0032, 0.22, 0.0032, 0.011	h^{-1}, –	[F]	Glycolytic/acidotic TOX^+ commitment vs. remodeling [5]
Perfusate pH / Flush	$k_{\mathrm{perf},0}$, $k_{\mathrm{perf,fl}}$, $k_{\mathrm{perf,LP}}$, $k_{\mathrm{fl,LP}}$, $\mathrm{pH}_{\mathrm{media}}$, k_{acid}, R_{buf}, N_{ref}	0.48, 1.80, 0.12, 0.85, 7.36, 0.007, 12.0, 48.0	h^{-1}, pH, fold	[A]	Hollow-fiber buffer exchange & bypass neutral flush

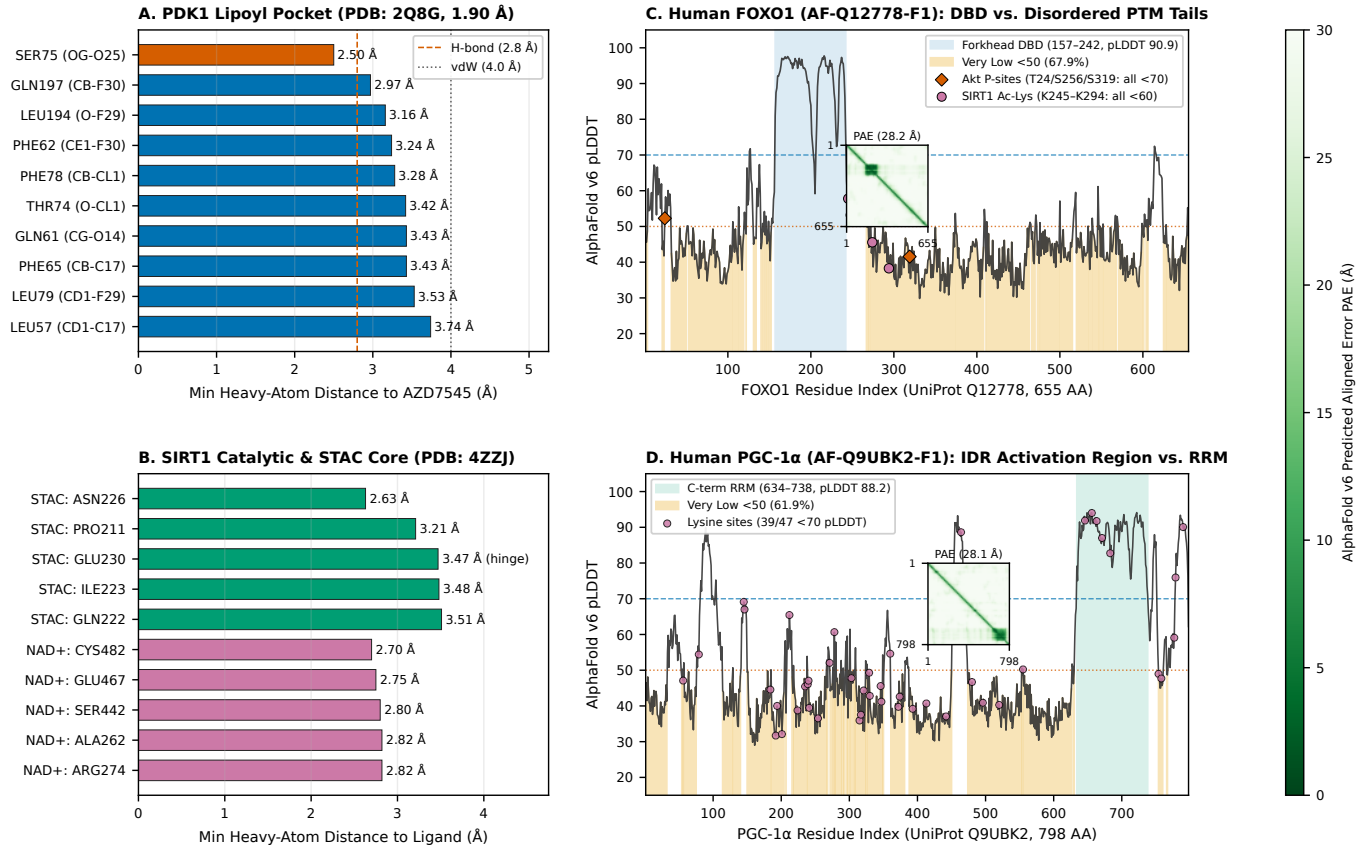


Figure 1: **Atomic Ligand-Contact and AlphaFold v6 pLDDT/PAE Structural Bioinformatics Analysis Guiding Stage-2 Actuator Selection. (A)** Heavy-atom contacts (≤ 4.0 Å) between human `PDK1` and the allosteric inhibitor `AZD7545` (`AZX`) in `PDB: 2Q8G` (1.90 Å X-ray [10]), anchored by a 2.50 Å hydrogen bond at `Ser75` (Oγ–O25) inside the N-terminal lipoyl-binding (`L2`) pocket. **(B)** Heavy-atom contacts (≤ 4.0 Å) in human `SIRT1` (`PDB: 4ZZJ`, 2.74 Å X-ray [15]) to the Sirtuin-Activating Compound `4TQ` (`Asn226` 2.63 Å; allosteric hinge `Glu230` 3.47 Å) and active-site carba-NAD^+ (`CNA`). **(C)** Full-length human `FOXO1` (`AF-Q12778-F1-v6`, 655 AA): only the Forkhead DNA-binding domain (`residues 157-242`, mean pLDDT = 90.89) is structured, whereas 9/9 `Akt` phosphorylation (`Thr24, Ser256, Ser319`) and `SIRT1` deacetylation (`Lys245-Lys294`) sites reside in disordered segments (pLDDT < 70; inset: 655×655 `PAE` matrix, off-diagonal mean 28.23 Å). **(D)** Full-length human `PGC-1`α (`AF-Q9UBK2-F1-v6`, 798 AA): only the C-terminal RNA-recognition motif (`residues 634-738`, mean pLDDT = 88.22) is structured, whereas 39/47 (82.98%) of lysine residues and the entire N-terminal/central activation region (`1-633`) are disordered (pLDDT < 70).

ligand hydroxyl (O25), flanked by `Gln197` (2.97 Å), `Leu194` (3.16 Å), `Phe62` (3.24 Å), `Phe78` (3.28 Å), `Thr74` (3.42 Å), `Gln61` (3.43 Å), `Phe65` (3.43 Å), `Leu79` (3.53 Å), `Leu57` (3.74 Å), `Asp64` (3.92 Å), and `Leu201` (3.92 Å). By severing `PDK1` tethering to the dihydrolipoamide acetyltransferase (`E2`) core of the `PDH` complex, `AZD7545` unblocks pyruvate oxidation without competing against millimolar intramitochondrial ATP.

Similarly, coordinate audit of the engineered human SIRT1 construct bound to Sirtuin-Activating Compound 4TQ and active-site carba-NAD$^+$ (PDB: 4ZZJ, Chain A, 344 resolved residues within 183-660 at 2.74 Å resolution [15]) resolves two distinct structured binding sites (Figure 1B): (i) the N-terminal three-helix STAC-binding domain, where 4TQ (36 heavy atoms) contacts Asn226 (2.63 Å Nδ2–OAN hydrogen bond), Pro211 (3.21 Å), the critical electrostatic hinge residue Glu230 (3.47 Å Oε1–NAB), Ile223 (3.48 Å), and Gln222 (3.51 Å); and (ii) the Rossmann-fold NAD$^+$-binding pocket, where carba-NAD$^+$ (CNA, 44 heavy atoms) is coordinated by Cys482 (2.70 Å), Glu467 (2.75 Å), Ser442 (2.80 Å), Ala262 (2.82 Å), and Arg274 (2.82 Å).

Chemical-Probe Caveat for SIRT1 STACs: Early synthetic STACs such as SRT1720 were shown by Pacholec et al. [12] to exhibit fluorophore-dependent artifacts on TAMRA-labeled peptides and off-target pharmacology in biochemical panels, while Hubbard et al. [13] demonstrated that Glu230-mediated allosteric activation of native substrates requires bulky hydrophobic residues at positions +1/ + 6 C-terminal to the target acetyl-lysine. Consequently, in our model and postulated benchtop falsification protocol (§7), Stage-2 SIRT1 activation (u_{sirt}) primarily represents NAD$^+$ substrate repletion via nicotinamide riboside (NR, acting directly at the Cys482/Glu467/Arg274 Rossmann pocket), with synthetic STACs treated strictly as an optional tool-compound arm requiring orthogonal genetic (SIRT1 CRISPR knockout) and Glu230Lys point-mutant controls.

3.2 Domain Architecture and PTM-Tail Disorder of FOXO1 and PGC-1α

Full-length AlphaFold v6 structural audit reveals a sharp dichotomy between isolated nucleic-acid-binding domains and disordered post-translational modification (PTM) regions in both downstream effectors (Figure 1C–D, Table 2):

1. **Human FOXO1 (AF-Q12778-F1-v6, 655 AA):** Global mean pLDDT is 50.96, with 67.94% (445/655 AA) in the *Very Low* (< 50) band, 18.17% (119/655) in 50–70, and 13.89% (91/655) $\geq$ 70, accompanied by a high off-diagonal mean PAE = 28.23 Å (2.02% residue pairs < 5 Å). Contiguous domain segmentation ($\geq$ 25 AA with pLDDT $\geq$ 70) identifies *one* structured domain: the winged-helix Forkhead DNA-binding domain (residues 157-242, 86 AA, mean pLDDT = 90.89, internal PAE = 3.21 Å). Crucially, however, **all 9/9 canonical regulatory PTM residues** controlling nuclear-cytoplasmic shuttling lie outside rigid tertiary pockets (pLDDT < 70): the three Akt 14-3-3 phosphosites (Thr24: 52.28, Ser256: 66.75, Ser319: 41.56) and all six CBP/p300/SIRT1 acetyl-lysines (Lys245: 55.34, Lys248: 53.38, Lys262: 57.78, Lys265: 52.34, Lys274: 45.38, Lys294: 38.28).

2. **Human PGC-1α (AF-Q9UBK2-F1-v6, 798 AA):** Global mean pLDDT is 52.74 (61.90% < 50, 17.17% in 50–70, 20.93% $\geq$ 70; off-diagonal mean PAE = 28.12 Å, 1.26% pairs < 5 Å). Segmentation detects *one* structured C-terminal RNA-recognition motif (RRM, residues 634-738, 105 AA, mean pLDDT = 88.22, internal PAE = 7.79 Å), whereas the entire 633-residue N-terminal and central transcriptional coactivation region—and 39/47 (82.98%) of all lysine residues across the protein (mean lysine pLDDT = 53.53)—is intrinsically disordered.

Rather than claiming that disordered transcription factors are universally "undruggable," this structural audit explains *mechanistically* why conventional pocket-based small-molecule discovery lacks preformed allosteric cavities in the PTM control regions of FOXO1 and PGC-1α, whereas the structured catalytic cleft of SIRT1 (PDB: 4ZZJ) is evolutionarily specialized to capture flexible acetyl-lysine peptide motifs.

4 In Silico Protocol Sweep, Target Ablations, and Adjoint Pareto Optimality

4.1 Nominal Switching Sweep and Mechanistic Component Ablations

To evaluate whether two-stage metabolic gating resolves the simulated proliferation–exhaustion conflict, we integrated the nominal 8-state system (RK4, N = 1,000 steps, Δt = 0.168 h) across 14 open-loop, closed-loop, and ablated regimens against five simultaneous manufacturing release criteria: viable expansion $N_{\text{cells}}(168\text{ h}) \geq 50.0\times$, stemness $\text{T}_{\text{SCM}}(168\text{ h}) \geq 68.0\%$, exhaustion $E_{\text{TOX}}(168\text{ h}) \leq 15.0\%$, SRC $\geq 3.5\times$, and depolarized organelle fraction $h(168\text{ h}) \leq 10.0\%$ (Figure 2A–B, Table 3).

Unphased Warburg expansion ($u_1 = u_2 = 0$) achieves maximal cell yield (81.77×) but drives severe mitochondrial depolarization (h = 66.76%), supernatant lactatemia (R_{LP} = 30.02, pH = 6.92), nuclear FOXO1 exclusion (F_{nuc} = 30.48%), TOX$^+$ commitment (76.13%), and T_{SCM} collapse (3.67%). Conversely, premature oxidative switching at Hour 0 ($u_1 = 0, u_2 = 1$) stalls clonal division at 9.58× (< 50× floor) while failing to clear baseline depolarized mitochondria (h = 64.81%). Continuous urolithin A monotherapy ($u_1 = 1, u_2 = 0$, without Stage-2 galactose/AZD7545/SIRT1 consolidation) thoroughly purges depolarized mitochondria (h = 0.04%) and preserves blastogenesis (80.60×), but because high glycolytic flux continues to drive Warburg exhaustion (41.03%) and prevents full cristae SRC maturation (2.26×), simulated T_{SCM} plateaus at 31.10%.

Table 2: **Verified Atomic-Coordinate and AlphaFold v6 Structural Bioinformatics Summary.** All metrics are computed directly from local `mmCIF` and `PAE` JSON coordinate files by `audit_all_structures()` in `simulator/generate_paper2_verified_benchmarks.py`.

Target Protein	Structure / Accession	Length / Res.	Structured Domain(s)	Disorder / PAE Metrics	Verified Key Contacts or PTM Sites
PDK1 (Kinase)	PDB: 2Q8G (X-ray)	368 AA (1.90 Å)	Lipoyl (L2) + Kinase core	Rigid co-crystal (41–423)	AZD7545: Ser75 (2.50 Å), Gln197 (2.97 Å), Leu194 (3.16 Å)
SIRT1 (Deacetylase)	PDB: 4ZZJ (X-ray)	344 AA (2.74 Å)	STAC 3-helix + Rossmann	Rigid co-crystal (183–660)	4TQ: Asn226 (2.63 Å), Glu230 (3.47 Å); CNA: Cys482 (2.70 Å)
FOXO1 (TF)	AF-Q12778-F1-v6	655 AA (AF v6)	Forkhead DBD 157–242 (90.9)	67.9% < 50; PAE = 28.23 Å	9/9 Akt (T24/S256/S319) & Ac-Lys (K245–K294) < 70 pLDDT
PGC-1α (Coact.)	AF-Q9UBK2-F1-v6	798 AA (AF v6)	C-term RRM 634–738 (88.2)	61.9% < 50; PAE = 28.12 Å	1–633 activation IDR; 39/47 (83.0%) lysines < 70 pLDDT

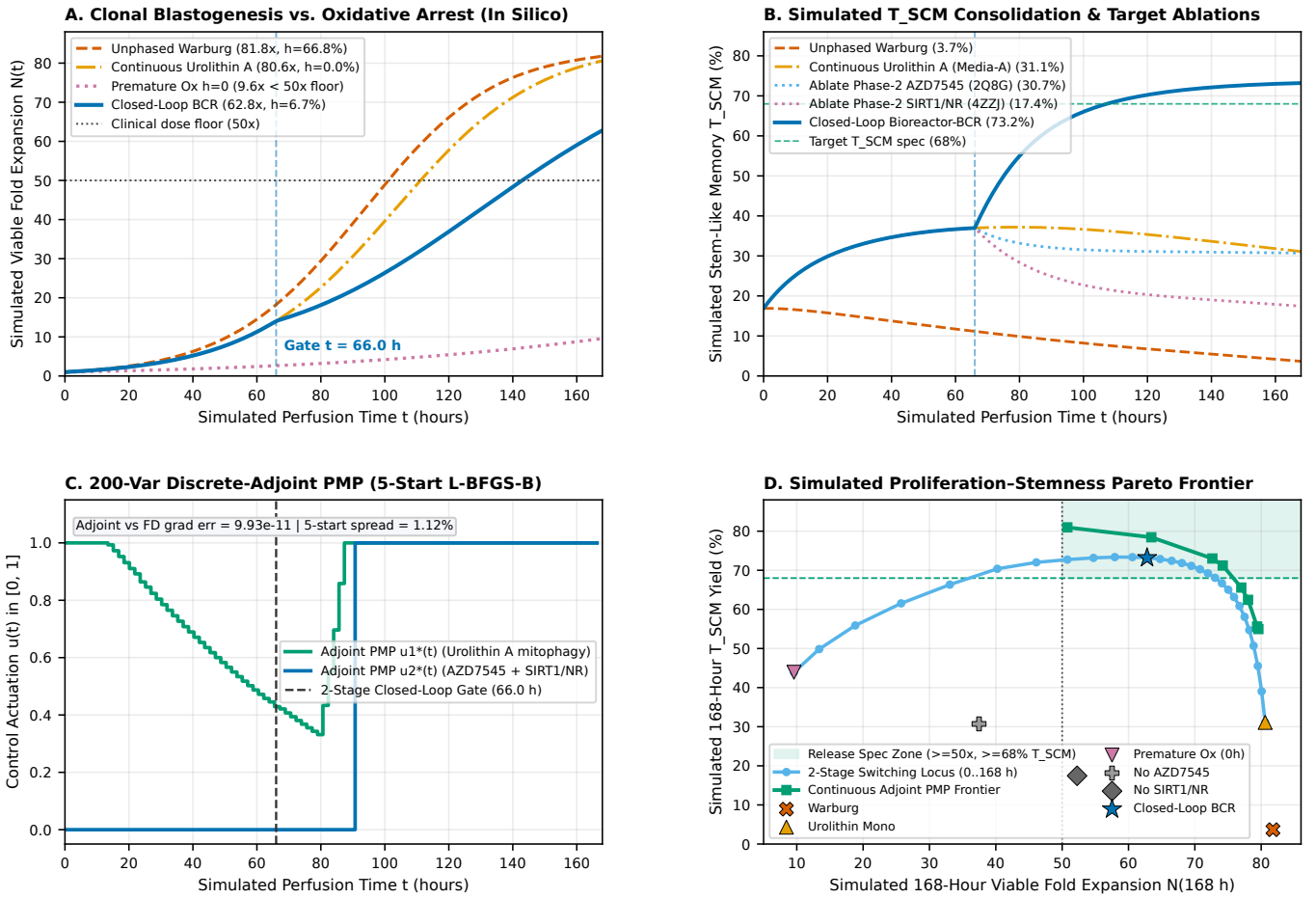


Figure 2: ***In Silico* 168-Hour Perfusion Trajectories, Component Ablations, and 200-Variable Discrete-Adjoint Pareto Frontier.** **(A)** Simulated clonal fold expansion $N_{\text{cells}}(t)$ showing Warburg glycolytic expansion (81.77×, h = 66.76%), premature Day-0 oxidative forcing (9.58× < 50× clinical floor), continuous urolithin A monotherapy (80.60×), and closed-loop **Bioreactor-BCR** switching at t_{gate} = 66.02 h (62.80×, h = 6.73%). **(B)** Simulated $\text{T}_{\text{SCM}}(t)$ consolidation comparing closed-loop gating (73.20%) against continuous urolithin A without Stage-2 oxidative consolidation (31.10%) and single-target Stage-2 ablations of `AZD7545` (30.72%) or `SIRT1/NR` (17.45%). **(C)** Unconstrained 200-variable discrete-adjoint Pontryagin optimal control schedule $(u_1^*(t), u_2^*(t)$ across 100 intervals; maximum adjoint vs. central finite-difference gradient error 9.93×10^{-11}, 5-start `L-BFGS-B` objective spread 1.12%). **(D)** Simulated proliferation–stemness Pareto frontier comparing continuous PMP modulation against the 2-stage switching locus ($t_{\text{sw}} \in [0, 168$ h]) and discrete baseline/ablation protocols.

Under nominal closed-loop gating ($h(t) \leq 0.045$, $N_{\text{cells}}(t) \geq 14.0\times$, $\text{pH}(t) \geq 7.10$), **Bioreactor-BCR** transitions from Media-A to Media-B at t_{gate} = 66.02 h (accumulated selection integral $s(t_{\text{gate}}) = 2.8248$), achieving 62.80× fold expansion, 73.20% T_{SCM}, SRC = 4.87×, E_{TOX} = 11.75%, and h(168 h) = 6.73%—satisfying all five release criteria simultaneously. Ablating Stage-2 `AZD7545` ($u_{\text{azd}} = 0$ after 66.02 h) leaves `PDH` phosphorylated ($g_{\text{pdh}} = 0.15$), cutting expansion in galactose-shifted media to 37.47× and T_{SCM} to 30.72%. Ablating Stage-2 `SIRT1/NR` activation ($u_{\text{sirt}} = 0$ after 66.02 h) allows D to out-replicate W ($k_{\text{rep,adv}} = 1.22$), causing h to rebound

from 4.50% at the gate to 36.27% at 168 h and collapsing T_{SCM} to 17.45%.

4.2 200-Variable Discrete-Adjoint Optimal Control and Pareto Frontier

To test whether a simple two-stage pinch-valve switch sacrifices substantial performance relative to an unconstrained, continuously modulated multi-pump schedule $(u_1(t), u_2(t)) \in [0,1]^2$, we discretized $[0, 168\ \text{h}]$ into $N_{ocp} = 100$ piecewise-constant control intervals (200 decision variables) and minimized the scalarized terminal-plus-effort Bolza objective

$$\begin{aligned} J_\alpha(\mathbf{u}) = &-\alpha \frac{T_{SCM}(\mathbf{x}(T))}{100} - (1-\alpha)\frac{N_{cells}(\mathbf{x}(T))}{N_{cap}} \\ &+ 0.45\, h(T) + 0.35\, E_{TOX}(T) \\ &+ 4.0\big(0.588 - N_{cells}(T)/N_{cap}\big)_+^2 \\ &+ \frac{0.008}{N_{ocp}} \sum_{k=0}^{N_{ocp}-1} \big(u_{1,k}^2 + u_{2,k}^2\big), \end{aligned} \tag{22}$$

propagating exact discrete-time `RK4` stage-by-stage Jacobians backward via the discrete costate recursion $\mathbf{p}_k = (\partial \Phi_k / \partial \mathbf{x}_k)^\top \mathbf{p}_{k+1}$ ($\mathbf{p}_k \in \mathbb{R}^8$). Across all 200 control variables, the maximum discrepancy between analytical discrete-adjoint gradients and central finite differences ($\epsilon = 10^{-6}$) is 9.93×10^{-11}. Launching `L-BFGS-B` from 5 diverse initializations (two-stage switch, uniform interior 0.5, linear cross-ramps, early 42 h switch, and pseudo-random interior schedule) converges to a tight objective spread of 1.12% ($J^*_{0.55} = -0.81366$, Table 3).

Interestingly, the Pontryagin-optimal control profile (Figure 2C) independently discovers a *two-phase temporal structure*: it holds $u_2^*(t) = 0$ throughout Hours 0–90.7 while tapering $u_1^*(t)$ along a smooth interior arc ($1.0 \to 0.33$, avoiding the small c_{akt} glycolytic proliferation penalty once $h(t)$ is low), executes a brief 15-hour co-actuation overlap ($u_1^* = 1, u_2^* = 0 \to 1$) to block $k_{rep,adv}$ rebound during initial mitochondrial biogenesis, and holds $u_1^* = u_2^* = 1$ to Hour 168. Sweeping the Pareto weight $\alpha \in [0.08, 0.92]$ (Figure 2D) shows that while continuous multi-pump modulation expands yield from 62.80× to 71.15× (+1.26% T_{SCM}), the single-valve two-stage locus already enters deep inside the 5-criteria clinical release zone without requiring continuous microfluidic drug mixing.

5 Heterogeneous Oncology Cohort, Optical Redox Observability, and Biofouling Handover

5.1 Why Supernatant `Lac:Pyr` Alone Fails Under Donor Heterogeneity and Electrode Fouling

A critical bioprocess engineering question is whether closed-loop feedback genuinely outperforms the best fixed open-loop switch timer across a heterogeneous patient population—and which non-destructive inline sensors are mathematically required for state observability. To test this without synthetic shortcuts, we generated a cohort of $N = 200$ simulated oncology apheresis donors with wide initial state dispersion ($h_0 \sim \mathcal{U}[0.20, 0.62]$, $M_0 \sim \mathcal{U}[0.85, 1.12]$, $R_{LP,0} \sim \mathcal{U}[18, 28]$, $F_{nuc,0} \sim \mathcal{U}[0.22, 0.36]$, $S_{SRC,0} \sim \mathcal{U}[1.05, 1.50]$, $E_{TOX,0} \sim \mathcal{U}[0.05, 0.14]$) and simultaneous unmodeled donor-to-donor kinetic perturbations ($\mu_{glyc} \times \mathcal{U}[0.66, 1.36]$, $\mu_{ox} \times \mathcal{U}[0.84, 1.16]$, $k_{mito,D,max} \times \mathcal{U}[0.68, 1.36]$, $k_{Warburg} \times \mathcal{U}[0.82, 1.22]$, and $k_{warb,exh} \times \mathcal{U}[0.82, 1.22]$). Crucially, the Extended Kalman Filter (EKF) runs strictly on `NOMINAL_PARAMS` and never observes any donor's private kinetic multipliers or initial state $\mathbf{x}(0)$.

We compare two non-destructive inline cassette sensor suites sampled every $\Delta t_{obs} = 0.30$ h (18 min) under correlated AR(1) noise ($\rho = 0.65$):

1. **4-Channel Supernatant/Respiration Suite (No `ORR`):** $y_1 = R_{LP} + \nu_1$ ($\sigma_1 = 0.55$, amperometric dual-enzyme electrode), $y_2 = \text{OUR}_{spec} + \nu_2 = \text{ATP}_{ox}(\mathbf{x}, u_2) + \nu_2$ ($\sigma_2 = 0.035$, optical O_2 optode), $y_3 = N_{cells} + \nu_3$ ($\sigma_3 = 0.45$, NIR backscatter/capacitance), and $y_4 = \text{pH} + \nu_4$ ($\sigma_4 = 0.018$, dual-wavelength phenol-red optical transmission).

2. **5-Channel Optical-Metabolic Suite (With Inline `ORR`):** Appends a fouling-immune, label-free quartz-window autofluorescence Optical Redox Ratio channel [14],

$$y_5 = \text{ORR}(\mathbf{x}, u_2) + \nu_5 = \frac{W}{W + D + c_{glyc}\, g_{glc}(u_2)} + \nu_5, \tag{23}$$

with $\sigma_5 = 0.012$, nominal $c_{glyc} = 0.35$, and ±8% unmodeled donor variation in c_{glyc} reflecting cytosolic NAD(P)H variation during aerobic glycolysis.

Computing the scaled empirical Observability Gramian $\mathcal{G} = \frac{1}{64}\int_0^{64} (\partial \tilde{\mathbf{y}}/\partial \tilde{\mathbf{x}}_0)^\top (\partial \tilde{\mathbf{y}}/\partial \tilde{\mathbf{x}}_0)\, dt$ across the six core gating states $[W, D, N_{cells}, R_{LP}, S_{SRC}, \text{pH}]$ exposes a stark structural observability bottleneck:

- Without `ORR`, the 4-channel Gramian has condition number $\kappa(\mathcal{G}_{4ch}) = 5{,}515.82$ (Jacobian condition 74.27). However, when protein biofouling disables amperometric channel y_1 (R_{LP}), the remaining

Table 3: **Nominal 168-Hour *In Silico* Protocol Sweep, Target Ablations, and Discrete-Adjoint PMP Benchmark.** Release specification: $N \geq 50\times$, $\mathrm{T_{SCM}} \geq 68\%$, $\mathrm{TOX^{+}} \leq 15\%$, $\mathrm{SRC} \geq 3.5\times$, and $h \leq 10\%$.

Simulated Regimen	t_{gate} (h)	$s(t_{\mathrm{gate}})$	**Fold** N(168 h)	$\mathrm{T_{SCM}}$ (%)	$\mathrm{TOX^{+}}$ (%)	SRC (×)	h(168 h) (%)	**5-Spec Pass**
Unphased Warburg ($u_1 = 0, u_2 = 0$)	–	–	81.77×	3.67%	76.13%	0.69×	66.76%	Fail (4/5)
Premature Oxidative ($u_1 = 0, u_2 = 1$ from 0 h)	0.00	0.0000	9.58×	43.92%	27.64%	1.22×	64.81%	Fail (5/5)
Continuous Urolithin A Monotherapy ($u_1 = 1, u_2 = 0$)	–	–	80.60×	31.10%	41.03%	2.26×	0.04%	Fail (3/5)
Fixed Early Switch ($t_{\mathrm{sw}} = 36$ h)	36.12	1.4696	46.11×	72.02%	9.32%	3.58×	23.51%	Fail (N, h)
Fixed Intermediate Switch ($t_{\mathrm{sw}} = 48$ h)	48.05	2.0009	54.72×	73.17%	10.16%	4.23×	14.87%	Fail ($h > 10\%$)
Closed-Loop Bioreactor-BCR ($h \leq 4.5\%$)	**66.02**	**2.8248**	**62.80×**	**73.20%**	**11.75%**	**4.87×**	**6.73%**	**PASS** (5/5)
Fixed Late Switch ($t_{\mathrm{sw}} = 96$ h)	96.10	4.2392	70.77×	70.26%	15.77%	5.13×	1.54%	Fail ($\mathrm{TOX^{+}}$)
Fixed Very Late Switch ($t_{\mathrm{sw}} = 120$ h)	120.12	5.3857	75.04×	65.03%	21.08%	4.78×	0.46%	Fail ($\mathrm{T_{SCM}}, \mathrm{TOX^{+}}$)
Ablate Stage-2 `AZD7545` (`PDB: 2Q8G`, $u_{\mathrm{azd}} = 0$)	66.02	2.8248	37.47×	30.72%	21.25%	2.66×	10.82%	Fail (5/5)
Ablate Stage-2 `SIRT1/NR` (`PDB: 4ZZJ`, $u_{\mathrm{sirt}} = 0$)	66.02	2.8248	52.25×	17.45%	28.46%	1.62×	36.27%	Fail (4/5)
Discrete-Adjoint PMP Optimal (200 vars, $\alpha = 0.55$)	Cont.	–	**71.15×**	**74.46%**	**14.08%**	**5.33×**	**0.09%**	**PASS** (5/5)

3-channel optical suite ($[\mathrm{OUR_{spec}}, N_{\mathrm{cells}}, \mathrm{pH}]$) senses mitochondrial mass almost exclusively through the linear combination $W + \eta_D D$, causing the Gramian condition number to explode 342× to $\kappa(\mathcal{G}_{\mathrm{3ch,foul}}) = 1{,}886{,}984.49$ (minimum singular value 1.49×10^{-5}). Moreover, even with an unfouled electrode, donor variation in k_{Warburg} (±22%) is structurally confounded with h in (14): a donor with naturally high glycolytic lactate efflux appears to the 4-channel EKF as having persistently high mitochondrial depolarization ($\hat{h}$ RMSE = 6.35%), delaying or preventing switching in 27/200 donors and yielding only a 37.5% **cohort pass rate**—worse than the best fixed 68 h open-loop timer (48.5%).

- Adding inline `ORR` (23) provides an orthogonal optical measurement of $W/(W + D + c_{\mathrm{glyc}} g_{\mathrm{glc}})$ whose sensitivity row $[\partial \mathrm{ORR}/\partial W, \partial \mathrm{ORR}/\partial D] = [(D + c_{\mathrm{glyc}} g_{\mathrm{glc}})/\Sigma^2, -W/\Sigma^2]$ is linearly independent of $[1, \eta_D]$ in $\mathrm{OUR_{spec}}$. Consequently, the 5-channel Gramian achieves $\kappa(\mathcal{G}_{\mathrm{5ch}}) = 4{,}906.11$, and after disabling fouled R_{LP}, the 4-channel all-optical post-fouling suite ($[\mathrm{OUR_{spec}}, N_{\mathrm{cells}}, \mathrm{pH}, \mathrm{ORR}]$) improves Gramian conditioning 12.6× over no-`ORR` (κ = 150,246.19 vs. 1,886,984.49), maintaining $\hat{h}$ estimation RMSE at 1.83% (unfouled) and 1.88% (biofouled).

5.2 Cohort Pass Rates, Feasibility Ceiling, and NIS Biofouling Handover

Across all $N = 200$ simulated oncology donors (Figure 3A–B, Table 4), unphased Warburg, premature Hour-0 oxidative switching, and continuous urolithin A monotherapy all record 0.0% (0/200) simultaneous 5-criteria pass rates. Sweeping nine open-loop fixed switch timers (24–120 h) reveals a narrow open-loop optimum at $t_{\mathrm{sw}} = 68$ h (48.5%, 97/200 donors) and 76 h (48.0%, 96/200); switching earlier (48 h : 26.0%) fails high-h_0 and hypoproliferative donors on $h > 10\%$ or $N < 50\times$, whereas switching later (96 h : 15.5%; 120 h : 0.0%) drives fast-dividing donors past the $\mathrm{TOX^{+}} \leq 15\%$ ceiling.

Evaluating an idealized full-information donor-specific oracle (exhaustively testing 28 candidate switch times per donor using private ground-truth states) establishes that 67.5% (135/200) of the heterogeneous cohort is controllable within a fixed 168-hour window using a two-stage protocol. The remaining 32.5% (65/200) infeasible subset is characterized by the Simultaneous Severe Exhaustion + Hypoproliferation phenotype: median initial depolarization $h_0 = 50.61\%$ combined with depressed glycolytic division rate 0.767× nominal (versus $h_0 = 36.35\%$ and 1.018× nominal among controllable donors), indicating that severely compromised apheresis starting material requires an extended (> 168 h, e.g., 192–216 h) manufacturing horizon rather than fixed-duration harvest.

Combining the 5-channel `ORR` Huber-EKF ($\delta_H = 2.5\sigma$ innovation clipping) with online optical-density growth-rate reachability gating ($\hat{h} \leq 0.056$, $\hat{N}_{\mathrm{cells}} \geq N_{\mathrm{req}}(t, \hat{\mu}_{\mathrm{OD}})$, 1.2 h 4-frame persistence quorum) raises the closed-loop pass rate to 55.0% (110/200 **donors**), outperforming the best possible retrospective fixed timer (48.5%) by +6.5 percentage points (+13.4% relative yield) and capturing 81.5% (110/135) of all theoretically controllable donors without patient-specific calibration (Figure 3B).

To stress-test sensor degradation, we injected progressive enzymatic protein biofouling into 100% of amperometric `Lac:Pyr` electrodes starting at $t_{\mathrm{foul}} \sim \mathcal{U}[30, 48]$ h (positive baseline drift +0.42 ratio units/h plus 14% erratic $\pm[4.8, 9.5]$ spikes, Figure 3C). A naive supernatant `Lac:Pyr` threshold gate ($R_{\mathrm{LP}} \leq 14.0$) is blinded by the upward biofouling drift and collapses to 0.5% (1/200). Monitoring the Channel-1 Normalized Innovation Squared statistic $\mathrm{NIS_1}(k) = \nu_1(k)^2/S_1(k)$ and permanently disabling y_1 after 3 consecutive frames above $\mathrm{NIS_1} > 13.5$ ($> 3.67\sigma$) hands estimation cleanly to the 4-channel all-optical quartet (`OUR+OD+pH+ORR`) across 200/200 fouled runs, retaining a 54.0% (108/200) **pass rate** ($\hat{h}$ RMSE 1.88%, versus 12.0% without `ORR`).

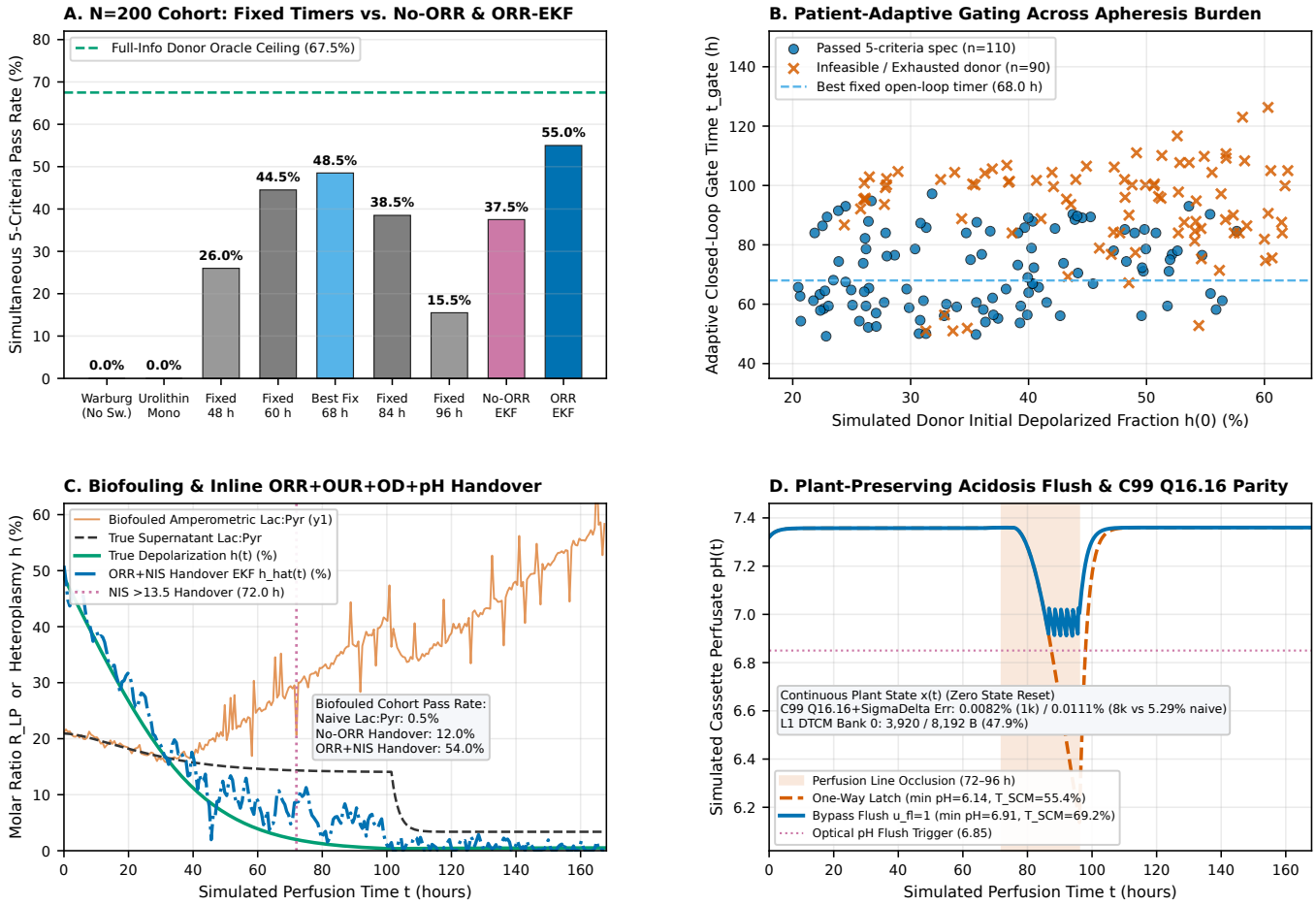


Figure 3: $N = 200$ **Heterogeneous Oncology Donor Cohort, Optical Redox (ORR) Observability Under Biofouling, and Plant-Preserving Acidosis Flush. (A)** Simultaneous 5-criteria manufacturing pass rates across $N = 200$ simulated oncology apheresis donors comparing unphased baselines (0.0%), nine open-loop fixed timers (peaking at 48.5% at 68 h), uncalibrated 4-channel EKF without ORR (37.5%), and the 5-channel ORR-augmented Huber-EKF (55.0%; full-information donor-specific oracle ceiling 67.5%). **(B)** Patient-adaptive closed-loop switching time t_{gate} versus donor initial mitochondrial depolarization $h_0 \in [20\%, 62\%]$ ($n = 110$ passing donors vs. $n = 90$ infeasible/hypoproliferative donors). **(C)** Representative biofouling trajectory (Donor #12) where enzymatic Lac:Pyr drift triggers NIS > 13.5 residual handover at 72.0 h to the fouling-immune optical quartet (OUR+OD+pH+ORR), preserving a 54.0% cohort pass rate (vs. 0.5% naive Lac:Pyr gate and 12.0% no-ORR handover). **(D)** Plant-preserving bypass neutral flush ($u_{\text{fl}} = 1$, zero discontinuous state reset) during simulated perfusion-line occlusion (72–96 h), lifting min pH from 6.14 to 6.91 and rescuing T_{SCM} from 55.39% to 69.16%.

6 Plant-Preserving Acidosis Rescue and Host-Emulated Firmware Verification

6.1 Continuous-Plant Neutral Bypass Flush Under Perfusion Occlusion

As established in Paper I [11], restoring a microcontroller's internal estimator memory after a fault cannot reset biological state variables; actuator recovery must operate strictly through continuous plant dynamics $\mathbf{x}(t)$. In hollow-fiber CAR-T cassettes, a primary post-switch hazard is partial peristaltic line occlusion or gas-transfer fouling during high-density Stage-2 expansion ($N_{\text{cells}} > 20\times$). We simulated a 24-hour Media-B perfusion-line occlusion across Hours 72–96 (92% drop in main-line exchange $k_{\text{perf},0}$ and $k_{\text{perf,LP}}$, loss of AZD7545 delivery $u_{\text{azd,eff}} = 0$, and hypoxic glycolytic surge $1.65 \times k_{\text{acid}}$, Figure 3D).

Under a one-way valve latch ($u_{\text{fl}} = 0$), cassette perfusate pH(t) plunges to min pH = 6.14, triggering severe acidotic FOXO1 nuclear export (15) and TOX^+ commitment (17) ($E_{\text{TOX}} = 32.28\%$, $\text{T}_{\text{SCM}} = 55.39\%$). When the supervisory controller monitors optical phenol-red pH(t) ≤ 6.85 and opens the auxiliary neutral-buffer bypass flush ($u_{\text{fl}} = 1$, reverting to protective Stage-1 media for a 6-hour stabilization window without altering $\mathbf{x}(t)$ discontinuously), min pH is arrested at 6.91, limiting

Table 4: **Simulated $N = 200$ Heterogeneous Oncology Apheresis Cohort and Amperometric `Lac:Pyr` Biofouling Benchmarks.** Medians and interquartile ranges (IQR) are reported across all 200 donors (MIASE PRNG seed `20261111`).

Cohort Control Policy ($N = 200$ Donors)	**Pass Rate**	**Fold N Med [IQR]**	**T_{SCM} Med [IQR]**	**TOX^+ Med**	**h(168 h) Med**	**$\hat{h}$ RMSE**	**Gate t_{sw} Med [IQR]**
Unphased Warburg ($u_1 = 0, u_2 = 0$)	0.0% (0/200)	79.7× [72.9, 83.5]	4.2% [2.1, 7.0]	74.6%	62.5%	–	–
Continuous Urolithin A Monotherapy	0.0% (0/200)	79.6× [69.7, 83.4]	30.6% [28.9, 32.4]	41.6%	0.04%	–	–
Open-Loop Fixed Timer (48 h)	26.0% (52/200)	50.7× [38.4, 65.3]	73.0% [71.1, 74.3]	10.8%	12.0%	–	48.0 h (fixed)
Open-Loop Fixed Timer (60 h)	44.5% (89/200)	57.1× [44.5, 69.5]	72.8% [71.5, 74.0]	11.8%	6.9%	–	60.0 h (fixed)
Best Open-Loop Fixed Timer (68 h)	**48.5%** (97/200)	60.6× [47.0, 72.3]	72.5% [71.1, 73.5]	12.6%	5.0%	–	68.0 h (fixed)
Open-Loop Fixed Timer (84 h)	38.5% (77/200)	65.8× [52.7, 75.9]	71.1% [69.9, 72.2]	14.5%	2.4%	–	84.0 h (fixed)
Open-Loop Fixed Timer (96 h)	15.5% (31/200)	68.8× [55.6, 77.8]	69.6% [68.3, 70.9]	16.4%	1.4%	–	96.0 h (fixed)
Ablation: 4-Ch Huber-EKF (**No `ORR`**, Clean)	37.5% (75/200)	66.2× [55.7, 75.2]	70.5% [66.2, 72.4]	15.0%	2.35%	6.35%	86.1 h [66.9, 100.5]
Closed-Loop 5-Ch `ORR` Huber-EKF (Clean)	**55.0%** (110/200)	63.9× [55.0, 71.4]	71.6% [69.6, 72.7]	13.8%	4.24%	**1.83%**	84.0 h [65.1, 95.2]
Biofouled `Lac:Pyr`: Naive Threshold Gate	0.5% (1/200)	79.5× [69.7, 83.4]	30.6% [28.9, 32.4]	41.6%	0.04%	46.87%	51.3 h ($n = 1$)
Biofouled `Lac:Pyr`: No-`ORR` NIS Handover	12.0% (24/200)	72.9× [58.0, 79.9]	65.3% [59.5, 69.3]	21.1%	0.52%	16.69%	114.3 h [100.2, 132.6]
Biofouled `Lac:Pyr`: `ORR` + NIS Handover	**54.0%** (108/200)	64.3× [54.8, 71.7]	71.5% [69.7, 72.7]	13.8%	4.05%	**1.88%**	84.0 h [66.3, 95.1]
Full-Information Donor Oracle Ceiling (168 h)	67.5% (135/200)	Upper bound of 2-stage protocols (32.5% hypoproliferative/high-h_0 donors need > 168 h)					

E_{TOX} to 16.11% and rescuing simulated T_{SCM} to 69.16% ($N_{cells} = 70.59\times$, $h = 8.14\%$).

6.2 Bare-Metal C99 `Q16.16/Q8.24` ΣΔ Firmware on `CoralNPU`

To verify that the 8-state CAR-T cassette observer-controller executes deterministically on ultra-low-power disposable-cassette microcontroller/NPU silicon without hardware floating-point dependencies, we implemented the complete cassette plant, fixed-point `RK4` integrator, NIS biofouling monitor, and acidosis bypass-flush state machine in freestanding C99 (`build/coralnpu/cart_bioreactor_hil_testbench.c`, compiled under `-O2 -Wall -Wextra -Werror -std=c99`). Because cell fold expansion $N_{cells} \in [1, 85]$ and supernatant $R_{LP} \in [3, 35]$ exceed $[-2, +2)$, states are stored in signed 32-bit `Q16.16` fixed-point with dynamic pre-scaling ($N_{cells}/64$, $R_{LP}/32$, $S_{SRC}/8$, pH/8), while `RK4` slopes are evaluated in higher-precision `Q8.24` ($2^{-24} \approx 5.96 \times 10^{-8}$ resolution) and accumulated into `Q16.16` states using a 64-bit ΣΔ sub-LSB remainder register.

Table 5 summarizes host-emulated verification results against IEEE-754 `FP64`:

1. **Single-Bank L1 DTCM Residency:** Static `sizeof(coralnpu_cart_dtcm_bank0_workspace_t)` is exactly 3,920 B (47.85% of the 8,192 B Bank 0 budget on Google Coral NPU `RV32IMF_Zve32x`, preserving Paper I's 3,696 B core layout), allowing Banks 1–3 (24 KB) to remain power-gated in 18.0 μW deep sleep between 18-minute optical frames.

2. **Zero Deadband Stalling Across Step Sizes:** At $N = 1{,}000$ steps, `Q16.16`+ΣΔ matches `FP64` switching time to the exact step (66.024 h vs. 66.024 h), final expansion (62.7802× vs. 62.7979×), and T_{SCM} (73.1881% vs. 73.1989%). Sweeping $N \in \{500, 1{,}000, 2{,}000, 4{,}000, 8{,}000\}$ steps keeps normalized ℓ_2 trajectory error flat at 0.0078%–0.0114%, whereas naive `Q16.16` without ΣΔ remainder tracking drifts 478× worse (0.435% → 5.291%) as $\Delta t \to 0.021$ h.

Table 5: **Host-Emulated C99 Fixed-Point Verification on `CoralNPU (RV32IMF_Zve32x)`.** Exact metrics emitted by `cart_bioreactor_hil_testbench.c --json`.

Hardware / Numerical Metric	**IEEE-754 `FP64`**	**C99 `Q16.16`+`Q8.24` ΣΔ**
L1 DTCM Bank 0 Static Workspace	–	3,920 B/8,192 B (47.85%)
Nominal Gate Switch Time ($N = 1{,}000$)	66.024 h	66.024 h
Nominal Fold Expansion N(168 h)	62.7979×	62.7802×
Nominal T_{SCM}(168 h) / h(168 h)	73.1989% / 6.7257%	73.1881% / 6.7253%
Norm. ℓ_2 Error ($N = 500$ / 1,000 / 2,000)	Ref.	0.0085% / 0.0082% / 0.0078%
Norm. ℓ_2 Error ($N = 4{,}000$ / 8,000)	Ref.	0.0114% / 0.0111%
Naive `Q16.16` Error ($N = 1{,}000$ / 8,000)	Ref.	0.6412% / 5.2906% *(drift)*
Acidosis Fault min pH (No Flush / Flush)	6.1366 / 6.9106	6.1366 / 6.9106
Active Burst (500 MHz) / Sleep Leakage	–	6.352 μs (3.108 mW, 19.74 nJ) / 18 μW

3. **Cycle-Weighted Burst Telemetry:** A complete 8-state `RK4` + sequential Huber-EKF + optical LUT + pinch-valve supervisory burst requires 3,176 cycles (6.352 μs at 500 MHz), consuming 3.108 mW active power (19.74 nJ per burst).

7 Limitations, Postulated Ex Vivo Falsification Blueprint, and Reproducibility

7.1 Model Limitations and Modeling Assumptions

Several limitations bound the interpretation of these computational results:

1. **Absence of Wet-Lab Execution:** No primary human T cells, CAR lentiviral vectors, or bioreactor cassettes were experimentally cultured in this study. Rate constants tagged **[F]** in Table 1 represent phenomenologically calibrated couplings synthesized from separate immunological and structural reports [1, 3, 4, 7] rather than joint system-identification fits to a single time-series dataset.

2. **Deterministic Population-Mean Approximation:** Equations (11)–(18) track mean intracellular organelle mass and population-level T_{SCM}/TOX^+

fractions, neglecting clone-to-clone CAR copy-number variance, $CD4^+/CD8^+$ subset interactions, and asymmetric mitochondrial partitioning at cytokinesis.

3. **Optical Redox Confounding:** While autofluorescence FLIM/intensity `ORR` achieves high accuracy in isolated T-cell activation assays [14], high-density hollow-fiber suspension scattering ($> 3 \times 10^7$ cells/mL) and riboflavin/phenol-red media background require spectral-window blanking and donor calibration in physical hardware.

7.2 Postulated 7-Day Ex Vivo Benchtop Falsification Protocol

To enable direct experimental falsification without requiring *in vivo* animal studies, our simulations generate four concrete, quantitative predictions testable in a standard 5-arm parallel perfusion bioreactor study using primary human CD19-BBζ CAR-T cells from $n \geq 12$ pretreated oncology donor apheresis samples:

- **Arm 1 (Standard Warburg Control):** 25 mM glucose + 300 IU/mL IL-2 (0–168 h). *Prediction:* High expansion ($> 70\times$), high $\text{TMRE}^{\text{low}}/\text{MitoTracker}^{\text{high}}$ depolarized fraction ($> 50\%$), $T_{SCM} < 10\%$.
- **Arm 2 (Continuous Urolithin A Monotherapy):** Media-A (25 μM urolithin A) continuously (0–168 h) in 25 mM glucose. *Prediction:* Clearance of TMRE^{low} mitochondria ($< 5\%$) but incomplete T_{SCM} conversion ($\sim$ 25–35%) due to persistent glycolytic TOX^+ acquisition.
- **Arm 3 (Fixed 68 h Open-Loop Switch):** Media-A (0–68 h) switched blindly at 68.0 h to Media-B (6.5 mM glucose + 18.5 mM galactose, 1.0 μM `AZD7545`, 1.0 mM nicotinamide riboside, 10 ng/mL IL-7/IL-15).
- **Arm 4 (Chemical-Probe-Free Metabolic + `AZD7545/NR` Closed-Loop Arm):** Closed-loop `ORR+OUR+OD+pH` Huber-EKF switching from Media-A to STAC-free Media-B (`AZD7545` + `NR` only, avoiding `SRT1720` off-target risk [12]). *Falsification Criterion:* Closed-loop gating is falsified if Arm 4 fails to increase the proportion of oncology donor runs simultaneously meeting $N \geq 50\times$, $\text{CD45RA}^+\text{CD62L}^+\text{TCF7}^+ \geq 68\%$, and Seahorse SRC $\geq 3.5\times$ relative to Arm 3, or if $h_0 > 48\%$ hypoproliferative donors do not exhibit delayed optimal gate times (> 80 h) as predicted in Figure 3B.
- **Arm 5 (`SIRT1-KO` / `Glu230Lys` & Single-Target Ablation Controls):** Closed-loop switching omitting either `AZD7545` or `NR` (testing the single-target ablation collapse predicted in Table 3), plus `SIRT1` CRISPR knockout / `Glu230Lys` point-mutant specificity controls if synthetic STACs (`4TQ/SRT2104`) are evaluated [13].

7.3 Code and Reproducibility Availability (MIASE)

All structural audits, 8-state `FP64` simulations, 200-variable discrete-adjoint `L-BFGS-B` optimizations, Observability Gramians, $N = 200$ donor Monte Carlo runs (PRNG seed `20261111`), publication vector figures, and bare-metal C99 fixed-point verification benchmarks are deterministically regenerated from source via `python3 simulator/generate_paper2_verified_benchmarks.py` and `build/coralnpu/cart_bioreactor_hil_testbench.c` at https://github.com/deesatzed/mitorestorer.

References

[1] Joseph A. Fraietta, Simon F. Lacey, Elena J. Orlando, Iulian Pruteanu-Malinici, Mercy Gohil, Stefan Lundh, Alina C. Boesteanu, Yan Wang, Roddy S. O'Connor, Wei-Ting Hwang, et al. Determinants of response and resistance to CD19 chimeric antigen receptor (CAR) T cell therapy of chronic lymphocytic leukemia. *Nature Medicine*, 24(5):563–571, 2018.

[2] Michael D. Buck, David O'Sullivan, Ramon I. Klein Geltink, Jonathan D. Curtis, Chih-Hao Chang, David E. Sanin, Jing Qiu, Oliver Kretz, Daniel Braas, Gerritje J. W. van der Windt, et al. Mitochondrial dynamics controls T cell fate through metabolic programming. *Cell*, 166(1):63–76, 2016.

[3] Alexander E. Doan, K. P. Mueller, A. Y. Chen, Crystal L. Mackall, et al. FOXO1 is a master regulator of memory programming in CAR T cells. *Nature*, 629(8010):211–218, 2024.

[4] Jack D. Chan, Christina M. Scheffler, Isabelle Munoz, Kevin Sek, Junyun Lee, Yu-Kuan Huang, Kah Min Yap, Vanessa T. Chew, Paul A. Beavis, Phillip K. Darcy, et al. FOXO1 enhances CAR T cell stemness, metabolic fitness and efficacy. *Nature*, 629(8010):201–210, 2024.

[5] Kristen E. Pauken, Morgan A. Sammons, Pamela M. Odorizzi, Sasikanth Manne, Jernej Godec, Omar Khan, Adam M. Drake, Zeyu Chen, Debattama R. Sen, Makoto Kurachi, R. Alan Barnitz, Caroline Bartman, Bertram Bengsch, Alexander C. Huang, Jason M. Schenkel, Golnaz Vahedi, W. Nicholas Haining, Shelley L. Berger, and E. John Wherry. Epigenetic stability of exhausted T cells limits durability of reinvigoration by PD-1 blockade. *Science*, 354(6316):1160–1165, 2016.

[6] Dongryeol Ryu, Laurent Mouchiroud, Pénélope A. Andreux, Elena Katsyuba, Norman Moullan, Amandine A. Nicolet-Dit-Félix, Evan G. Williams, Prashant Jha, Giuseppe Lo Sasso, Chris Rinsch, Johan Auwerx, et al. Urolithin A induces mitophagy and prolongs lifespan in *C. elegans* and increases muscle function in rodents. *Nature Medicine*, 22(8):879–888, 2016.

[7] Dominic Denk, Valentina Petrocelli, Claire Conche, Moritz Drachsler, Paul K. Ziegler, Adele Braun, Julia Körner, Michael Kasper, Chris Rinsch, Florian R. Greten, et al. Expansion of T memory stem cells with superior anti-tumor immunity by Urolithin A-induced mitophagy. *Immunity*, 55(11):2059–2073, 2022.

[8] Dominic Denk, Anurag Singh, Herbert G. Kasler, Davide D'Amico, Chris Rinsch, Eric Verdin, Florian R. Greten, et al. Effect of the mitophagy inducer urolithin A on age-related immune decline: a randomized, placebo-controlled trial. *Nature Aging*, 5(11):2309–2322, 2025.

[9] Madhusudhanan Sukumar, Jie Liu, Yun Ji, Medhavi Subramanian, Joseph G. Crompton, Zhiya Yu, Rahul Roychoudhuri, Douglas C. Palmer, Pawel Muranski, Edward D. Karoly, et al. Inhibiting glycolytic metabolism enhances CD8+ T cell memory and antitumor function. *Journal of Clinical Investigation*, 123(10):4479–4488, 2013.

[10] Masato Kato, Jun Li, Jacinta L. Chuang, and David T. Chuang. Distinct structural mechanisms for inhibition of pyruvate dehydrogenase kinase isoforms by AZD7545, dichloroacetate, and radicicol. *Structure*, 15(8):992–1004, 2007.

[11] O. Satz. Nondimensional staged feedback control of mitochondrial heteroplasmy dynamics with host-emulated verification on an ultra-low-power RV32IMF_Zve32x neural processing unit. MitoRestorer Series, Paper I; https://github.com/deesatzed/mitorestorer, 2026.

[12] Michelle Pacholec, John E. Bleasdale, Boris Chrunyk, David Cunningham, Declan Flynn, Robert S. Garofalo, David Griffith, Matthew Griffor, Pat Loulakis, Brian Pabst, et al. SRT1720, SRT2183, SRT1460, and resveratrol are not direct activators of SIRT1. *Journal of Biological Chemistry*, 285(11):8340–8351, 2010.

[13] Basil P. Hubbard, Ana P. Gomes, Han Dai, Jun Li, April W. Case, Thomas Considine, Thomas V. Riera, Jessica E. Lee, Sook Yen E, Dudley W. Lamming, et al. Evidence for a common mechanism of SIRT1 regulation by allosteric activators. *Science*, 339(6124):1216–1219, 2013.

[14] Alex J. Walsh, Katherine P. Mueller, Kelsey Tweed, Isabel Jones, Cole M. Walsh, Nicole J. Piscopo, Natalie M. Niemi, David J. Pagliarini, Krishanu Saha, and Melissa C. Skala. Classification of T-cell activation via autofluorescence lifetime imaging. *Nature Biomedical Engineering*, 5(1):77–88, 2021.

[15] Han Dai, April W. Case, Thomas V. Riera, Thomas Considine, Jessica E. Lee, Yoshitomo Hamuro, Huizhen Zhao, Yong Jiang, Sharon M. Sweitzer, Charles A. Blum, Jeremy S. Disch, Pui Yee Ng, Robert B. Perni, George P. Vlasuk, and James L. Ellis. Crystallographic structure of a small molecule SIRT1 activator-enzyme complex. *Nature Communications*, 6:7645, 2015.

[16] James B. Stewart and Patrick F. Chinnery. The dynamics of mitochondrial DNA heteroplasmy: implications for human health and disease. *Nature Reviews Genetics*, 16(9):530–542, 2015.

[17] John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873):583–589, 2021.

[18] Mihaly Varadi, Damiano Bertoni, Paulyna Magana, Urmila Paramval, Ivanna Pidruchna, Malarvizhi Radhakrishnan, Maxim Tsenkov, Sreenath Nair, Milot Mirdita, Jingi Yeo, et al. AlphaFold Protein Structure Database in 2024: providing structure coverage for over 214 million protein sequences. *Nucleic Acids Research*, 52(D1):D368–D375, 2024.